Fundamentals 21

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Problem 21.1

Let $\mathcal{T} = \{\varnothing, X\}$. We show that it satisfies the three axioms:

1. $\varnothing, X \in \mathcal{T}$.
2. $\varnothing \cup X = X \in \mathcal{T}$.
3. $\varnothing \cap X = \varnothing \in \mathcal{T}$.

Now, let $\mathcal{T} = \mathcal{P}(X)$.

1. $\varnothing$ and X are subsets of X so $\varnothing, X \in \mathcal{P}(X)$.
2. Assume $\{X_\alpha : \alpha \in A\} \subseteq \mathcal{T}$, then for any α and $x \in X_\alpha$, $x \in X$. So $\bigcup_{\alpha \in A} X_\alpha \in \mathcal{P}(X)$.
3. Let $Y, Z \in \mathcal{P}(X)$, so that Y and Z are subsets of X. We have that $Y \cap Z$ is a subset of X, so $Y \cap Z \in \mathcal{P}(X)$.

Problem 21.3

Since the evenly spaced integer topology is defined as the set of unions of all $\{a\mathbb{Z} + b\}$, it suffices to show that $\mathcal{B} = \{a\mathbb{Z} + b : a, b \in \mathbb{Z}, a \neq 0\}$ is a topological basis.

1. For any $x \in \mathbb{Z}$, $x \in \{1\mathbb{Z} + x\} \in \mathcal{B}$.
2. If $x \in a\mathbb{Z} + b$ and $c\mathbb{Z} + d$, then $x \in \operatorname{lcm}(a, c)\mathbb{Z} + x \subseteq (a\mathbb{Z} + b) \cap (c\mathbb{Z} + d)$.

Problem 21.4

First, we show that $\mathcal{T}$ is a topological basis:

1. For any $x \in X$, since $X \in \mathcal{T}$, $x \in X \in \mathcal{T}$.
2. If $x \in B_1 \cap B_2$, since $B_3 = B_1 \cap B_2 \in \mathcal{T}$, $x \in B_3 \subseteq B_1 \cap B_2$.

Since $\mathcal{T}$ is closed under arbitrary unions, all unions of elements in $\mathcal{T}$ are also in $\mathcal{T}$, so $\mathcal{T}$ can be generated by itself.

Problem 21.5

First assume that $\mathcal{B}$ is a basis for $\mathcal{T}$. Let $U \in \mathcal{T}$ and $x \in U$. We know that U is a union of elements in $\mathcal{B}$, say $\{B_\alpha : \alpha \in A\}$. Since $x \in U$, we have that $x \in B_\alpha$ for some $\alpha \in A$, and $B_\alpha \subseteq U$ because U is a union of B_α and other sets.

Now, assume that for each $U \in \mathcal{T}$ and $x \in U$ there's $B \in \mathcal{B}$ such that $x \in B \subseteq U$. We need to show that all elements of $\mathcal{T}$ are unions of elements in $\mathcal{B}$. For contradiction, assume this is false. Then there's $U \in \mathcal{T}$ where $U \neq \bigcup_{\alpha \in A} X_\alpha$ for any $\{X_\alpha : \alpha \in A\} \subseteq \mathcal{B}$. Then there's some $x \in U$ where $x \notin \mathcal{B}$ for any $B \in \mathcal{B}$, which is a contradiction.

Problem 21.6

Part (1)

1. Let $x \in \mathbb{R}$ and $B = (x-1, x+1)$. Then $B \in \mathcal{B}$ and $x \in B$.
2. Let $B_3 = B_1 \cap B_2$. Since B_3 is open, $B_3 \in \mathcal{B}$.

Part (2)

We first show that every open set in $\mathbb{R}$ is a union of open intervals. Let $U \in \mathbb{R}$ open. Then for all $x \in X$, there exists ε_x such that $B_\varepsilon(x) \subseteq U$. Therefore, $\bigcup_{x \in X} B_{\varepsilon_x}(x) \subseteq U$. Also, since x is in some $B_{\varepsilon_x}(x)$ for all $x \in U$, $U \subseteq \bigcup_{x \in X} B_{\varepsilon_x}(x)$. So we have that $U = \bigcup_{x \in X} B_{\varepsilon_x}(x)$.

Now, since all open sets can be written as arbitrary unions of elements in $\mathcal{B}$, the topology generated by $\mathcal{B}$ is just all sets that are open by the Euclidean metric.

Part (3)

Every open set is a union of open intervals (a, b), so it suffices to show that all open intervals are arbitrary unions of intervals (a, b) where $a, b \in \mathbb{Q}$. We can find rational sequences $(a_n), (b_n)$ such that $a = \lim_{n \to \infty} a_n$ and $b = \lim_{n \to \infty} b_n$, where $a_n > a$ and $b_n < b$ for all $n \in \mathbb{N}$. So we have that $(a, b) = \bigcup_{n \in \mathbb{N}} (a_n, b_n)$.

Problem 21.8

Let $\{X_\alpha : \alpha \in A\} \subseteq X$ be a set of neighborhoods of x. So there is some $U \in \mathcal{T}$ where $x \in U \subseteq X_\alpha$ for some $\alpha \in A$. Then $U \subseteq \bigcup_{\alpha \in A} X_\alpha$, so $\bigcup_{\alpha \in A} X_\alpha$ is a neighborhood containing X.

Let $Y, Z \subseteq X$ be neighborhoods of x. Then there are $U_1 \subseteq Y$, $U_2 \subseteq Z$ where $U_1, U_2 \in \mathcal{T}$. So $U_1 \cap U_2 \subseteq Y \cap Z$, and since $U_1 \cap U_2$ is open and $x \in U_1 \cap U_2$, $Y \cap Z$ is a neighborhood of x.

Problem 21.9

Let $T = \bigcap_{\alpha \in A} \mathcal{T}_\alpha$.

1. $\varnothing, X \in \mathcal{T}_\alpha$ for all $\alpha \in A$, s $\varnothing, X \in \mathcal{T}$.
2. Let B be any set. If $\{X_\beta : \beta \in B\} \subseteq \mathcal{T}$ then $\{X_\beta : \beta \in B\} \in \mathcal{T}_\alpha$ for all $\alpha \in A$. Since arbitrary unions of sets in $\mathcal{T}_\alpha$ are also in $\mathcal{T}_\alpha$ for all $\alpha \in A$, $\bigcup_{\beta \in B} X_\beta \in \mathcal{T}_\alpha$ for all $\alpha \in A$, so $\bigcup_{\beta \in B} X_\beta \in \mathcal{T}$.
3. Let $Y, Z \in \mathcal{T}$, so that $Y, Z \in \mathcal{T}_\alpha$ for all $\alpha \in A$. We have that $Y \cap Z \in T_\alpha$ for all $\alpha \in A$, so $Y \cap Z \in \mathcal{T}$.

Problem 21.10

We can find a counterexample. Let $\mathcal{T}$ be the topology induced by the Euclidean metric on $\mathbb{R}$ and $\mathcal{T}'$ be the topology induced by the discrete metric on $\mathbb{R}$. Open balls in $\mathcal{T}'$ are one element sets. So, for any $x \in \mathbb{R}$, the set $x \in T'$ is contained in $(x-1, x+1) \in \mathcal{T}$. Also, for any $U \in \mathcal{T}$, U is contained in the set $\bigcup_{x \in U} \{x\} \in T'$.

We have that the set $\{0, 1\} \in \mathcal{T}'$, but it is not open under the Euclidean metric, so $\{0, 1\} \notin \mathcal{T}$.